

COMBINATORICS

EXAM 1

ROB TOMPKINS

1. (25pts): Let a_n be the number of ways to partition an n -element set into ordered pairs (i.e. matchings where order within each pair matters). Find the exponential generating function for a_n . Derive a closed-form expression for a_n .

Solution.

□

2. (25pts): Let a_n be the number of ways to distribute n labeled objects into any number of labeled boxes such that each box contains at least two objects. Construct the exponential generating function for a_n . Extract a formula for a_n .

Solution.

□

3. (25pts): The Catalan numbers are defined by $C_n = \frac{1}{1+n} \binom{2n}{n}$. Prove that C_n satisfies the recurrence $C_n = \sum_{k=0}^{n-1} C_k C_{n-1-k}$. Give a combinatorial interpretation of this recurrence.

Proof.

□

4. (25 pts): How many integers between 1 and 1000 (inclusive) are divisible by at least one of 3, 5, or 7? Use the Inclusion-Exclusion Principle and show all steps.

Solution.

□

5. (25pts): Suppose we have an infinite supply of pennies and we would like to stack them up in rows. Find the number of different arrangements possible subject to the following conditions:

- (i) the bottom row consists of n pennies, each touching its two neighbors, except the ends where $n \geq 1$;
- (ii) a coin that does not belong to a row sits on two coins below it;
- (iii) do not distinguish between heads and tails.

Solution.

□

6. (25pts): Consider the colorings of the vertices of a regular hexagon using 3 colors. Determine the cycle index polynomial of the dihedral group D_6 . Use Polya's Enumeration Theorem to compute the number of distinct colorings.

Solution.

□

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7. (25pts): Model a benzene ring as a hexagon. Suppose we substitute hydrogen atoms with two types of atoms (A and B). Using Polya's Enumeration Theorem, determine the number of distinct substitution patterns (up to rotational and reflectional symmetry).

Solution.

□

8. (25pts): Show that: $R(3, 5) > 13$ and $R(3, 5) \leq 14$.

Proof.

□